

# QIC 890: Quantum Complexity Theory

## Course Project

Due 11:59 pm Eastern Time, Friday, December 11, 2026

### Project description

The course project is a written document on a topic in quantum complexity theory. You may view this document as a chapter of lecture notes for a hypothetical lecture on the chosen topic, along the lines of the course lecture notes. It should be self-contained within the context of this course, meaning that any required definitions and concepts not covered in the course lecture notes should be properly introduced and explained.

- Some projects may focus on one main theorem. Required definitions and background information should be included along with a clear statement of the theorem and presentation of its proof.
- Complete proofs of every result are not required. In particular, difficult-to-prove results from other sources can, at your discretion, be cited rather than proved.
- Other topics may be more open-ended, such as a survey of an area. In such cases, the goal is not to cover everything — but rather to present results that you find most interesting or view as important.

### Format and length

- Reports must be typeset in  $\text{\LaTeX}$ . A template using the same page layout, font, and macros as the course notes has been posted to the course website for convenience (but feel free to use a different style).
- There is no rigid length requirement. The suggested length is that of a typical chapter in the course lecture notes — generally 12–24 pages in the template’s format (12-point font on a  $5.5 \times 8.5$ -inch text block).

## Target audience and level of detail

Write for a hypothetical classmate who has read and understood the course notes, but who knows nothing about your topic beyond that. The course notes are a reasonable target for the level of detail that is appropriate.

## Sources and AI

- Use of AI tools is allowed provided you honestly acknowledge this. However, the report you submit must be written by you, in your own words. Do not ask AI to write your project for you!
- Anything other than the course notes must be cited, including books, papers, lecture notes from other courses, and websites. Facts proved in the notes do not need to be proved again and can be referred to as needed.
- The references listed with each topic below are starting points, not restrictions. Naturally, you may find other sources or follow-up work to be important for your project.

## Logistics

- Email your report as a single PDF attachment to [watrous@uwaterloo.ca](mailto:watrous@uwaterloo.ca), with the subject “QIC 890 project”.
- Reports are due at 11:59 pm Eastern Time on Friday, December 11, 2026.
- You are welcome to discuss with me your choice of topic or project more generally at any stage.

## Suggested topics

Please feel free to ask for clarifications about any of these topics. Chapter references are to the course notes.

1. *Quantum Turing machines and quantum circuits*. The course is based around the quantum circuit model, but quantum Turing machines came first. Possible

projects include the equivalence of quantum Turing machines and uniform families of quantum circuits, and the construction of an efficient universal quantum Turing machine. Background: Chapters 1 and 4.

- E. Bernstein and U. Vazirani. Quantum complexity theory. *SIAM Journal on Computing* 26(5): 1411–1473, 1997.
- A. C.-C. Yao. Quantum circuit complexity. In *Proceedings of the 34th Annual Symposium on Foundations of Computer Science*, pages 352–361, 1993.
- A. Molina and J. Watrous. Revisiting the simulation of quantum Turing machines by quantum circuits. *Proceedings of the Royal Society A* 475(2226): 20180767, 2019. arXiv:1808.01701.

2. *The Solovay–Kitaev theorem.* Chapter 4 states the Solovay–Kitaev theorem without proof. Possible projects include a proof of the theorem, lower bounds on the length of approximating circuits, and a survey of more recent refinements. Background: Chapter 4.

- C. M. Dawson and M. A. Nielsen. The Solovay–Kitaev algorithm. *Quantum Information and Computation* 6(1): 81–95, 2006. arXiv:quant-ph/0505030.
- A. Yu. Kitaev, A. H. Shen, and M. N. Vyalyi. *Classical and Quantum Computation*. Graduate Studies in Mathematics, volume 47. American Mathematical Society, 2002.
- A. W. Harrow, B. Recht, and I. L. Chuang. Efficient discrete approximations of quantum gates. *Journal of Mathematical Physics* 43(9): 4445–4451, 2002. arXiv:quant-ph/0111031.
- G. Kuperberg. Breaking the cubic barrier in the Solovay–Kitaev algorithm. arXiv:2306.13158, 2023.

3. *Exact synthesis.* Chapter 4 also states, without proof, a characterization of the unitary matrices that can be implemented exactly by circuits of Hadamard, CNOT, and  $T$  gates. Possible projects include a proof of this characterization, as well as analogous characterizations for other gate sets. Background: Chapters 4 and 6.

- V. Kliuchnikov, D. Maslov, and M. Mosca. Fast and efficient exact synthesis of single-qubit unitaries generated by Clifford and  $T$  gates. *Quantum Information and Computation* 13(7–8): 607–630, 2013. arXiv:1206.5236.
- B. Giles and P. Selinger. Exact synthesis of multiqubit Clifford+ $T$  circuits. *Physical Review A* 87(3): 032332, 2013. arXiv:1212.0506.

- M. Amy, A. N. Glaudell, and N. J. Ross. Number-theoretic characterizations of some restricted Clifford+T circuits. *Quantum* 4: 252, 2020. arXiv:1908.06076.
  - M. Amy, A. N. Glaudell, S. Kelso, W. Maxwell, S. S. Mendelson, and N. J. Ross. Exact synthesis of multiqubit Clifford-cyclotomic circuits. In *Reversible Computation (RC 2024)*, Lecture Notes in Computer Science 14680, pages 238–245, 2024. arXiv:2311.07741.
4. *Gate sets with algebraic amplitudes.* The proof that  $BQP \subseteq PP$  in Chapter 6 relies on the entries of the standard gates being Gaussian integers (after scaling). The containment holds for any finite gate set with algebraic entries, as Adleman, DeMarrais, and Huang showed. A possible project is to obtain this more general result through GapP functions, along the lines of Chapter 6. Background: Chapter 6 (and some basic algebraic number theory).
- L. M. Adleman, J. DeMarrais, and M.-D. A. Huang. Quantum computability. *SIAM Journal on Computing* 26(5): 1524–1540, 1997.
  - L. Fortnow and J. Rogers. Complexity limitations on quantum computation. *Journal of Computer and System Sciences* 59(2): 240–252, 1999. arXiv:cs/9811023.
5. *Classical simulation of restricted quantum circuits.* Some families of quantum circuits can be simulated efficiently by classical computers, including stabilizer (Clifford) circuits and matchgate circuits, which correspond to non-interacting fermions. A project could survey one or more such families and how they can be simulated classically — and perhaps what needs to be added to recover universal quantum computation. Background: Chapters 4 and 5.
- D. Gottesman. The Heisenberg representation of quantum computers. In *Group22: Proceedings of the XXII International Colloquium on Group Theoretical Methods in Physics*, pages 32–43. International Press, 1999. arXiv:quant-ph/9807006.
  - S. Aaronson and D. Gottesman. Improved simulation of stabilizer circuits. *Physical Review A* 70(5): 052328, 2004. arXiv:quant-ph/0406196.
  - S. Bravyi and A. Kitaev. Universal quantum computation with ideal Clifford gates and noisy ancillas. *Physical Review A* 71(2): 022316, 2005. arXiv:quant-ph/0403025.
  - L. G. Valiant. Quantum circuits that can be simulated classically in polynomial time. *SIAM Journal on Computing* 31(4): 1229–1254, 2002.
  - B. M. Terhal and D. P. DiVincenzo. Classical simulation of noninteracting-fermion quantum circuits. *Physical Review A* 65(3): 032325, 2002. arXiv:quant-ph/0108010.

- R. Jozsa and A. Miyake. Matchgates and classical simulation of quantum circuits. *Proceedings of the Royal Society A* 464(2100): 3089–3106, 2008. arXiv:0804.4050.
6. *Unconditional separations for shallow circuits.* Constant-depth quantum circuits can solve certain relation problems that classical circuits of constant depth and bounded fan-in cannot — which is a rare example of an unconditional separation between quantum and classical computation. Possible projects include this result and its extension to classical circuits with unbounded fan-in. Background: Chapters 3 and 4.
- S. Bravyi, D. Gosset, and R. König. Quantum advantage with shallow circuits. *Science* 362(6412): 308–311, 2018. arXiv:1704.00690.
  - A. Bene Watts, R. Kothari, L. Schaeffer, and A. Tal. Exponential separation between shallow quantum circuits and unbounded fan-in shallow classical circuits. In *Proceedings of the 51st Annual ACM Symposium on Theory of Computing*, pages 515–526, 2019. arXiv:1906.08890.
7. *Hamiltonian simulation is BQP-complete.* Simulating the evolution of a quantum system under a local Hamiltonian is, in a suitable formulation, a BQP-complete problem. Possible projects include a proof of this fact and a survey of techniques for Hamiltonian simulation and their error analyses. Background: Chapters 4, 5, and 10.
- S. Lloyd. Universal quantum simulators. *Science* 273(5278): 1073–1078, 1996.
  - R. P. Feynman. Quantum mechanical computers. *Foundations of Physics* 16(6): 507–531, 1986.
  - D. W. Berry, G. Ahokas, R. Cleve, and B. C. Sanders. Efficient quantum algorithms for simulating sparse Hamiltonians. *Communications in Mathematical Physics* 270(2): 359–371, 2007. arXiv:quant-ph/0508139.
  - D. Nagaj. Fast universal quantum computation with railroad-switch local Hamiltonians. *Journal of Mathematical Physics* 51(6): 062201, 2010. arXiv:0908.4219.
  - A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu. Theory of Trotter error with commutator scaling. *Physical Review X* 11(1): 011020, 2021. arXiv:1912.08854.
8. *Complexity foundations of quantum advantage experiments.* Experiments aimed at demonstrating quantum advantage are often based on sampling, where a quantum device produces samples from some output distribution (e.g., running a randomly generated quantum circuit and measuring, or recording photon detector outputs in a linear-optical network). Complexity-theoretic

arguments show, or in some cases suggest through unproved conjectures, that performing this sampling with a classical computer is hard. A project could explain these arguments or survey work on this topic. Background: Chapter 7.

- M. J. Bremner, R. Jozsa, and D. J. Shepherd. Classical simulation of commuting quantum computations implies collapse of the polynomial hierarchy. *Proceedings of the Royal Society A* 467(2126): 459–472, 2011. arXiv:1005.1407.
  - S. Aaronson and A. Arkhipov. The computational complexity of linear optics. *Theory of Computing* 9(4): 143–252, 2013. arXiv:1011.3245.
  - M. J. Bremner, A. Montanaro, and D. J. Shepherd. Average-case complexity versus approximate simulation of commuting quantum computations. *Physical Review Letters* 117(8): 080501, 2016. arXiv:1504.07999.
  - S. Aaronson and L. Chen. Complexity-theoretic foundations of quantum supremacy experiments. In *Proceedings of the 32nd Computational Complexity Conference, LIPIcs* 79, pages 22:1–22:67, 2017. arXiv:1612.05903.
  - A. Bouland, B. Fefferman, C. Nirkhe, and U. Vazirani. On the complexity and verification of quantum random circuit sampling. *Nature Physics* 15(2): 159–163, 2019. arXiv:1803.04402.
9. *Eliminating intermediate measurements in quantum logspace.* Deferring measurements to the end of a computation requires storing qubits, which contributes to space usage — but it is known that intermediate measurements can be eliminated from space-bounded quantum computations without a significant increase in space. Possible projects include a proof of this result, or related results on well-conditioned linear-algebraic problems in quantum logspace. Background: Chapters 2, 4, and 5.
- B. Fefferman and Z. Remscrem. Eliminating intermediate measurements in space-bounded quantum computation. In *Proceedings of the 53rd Annual ACM Symposium on Theory of Computing*, pages 1343–1356, 2021. arXiv:2006.03530.
  - U. Girish, R. Raz, and W. Zhan. Quantum logspace algorithm for powering matrices with bounded norm. In *Proceedings of the 48th International Colloquium on Automata, Languages, and Programming, LIPIcs* 198, pages 73:1–73:20, 2021. arXiv:2006.04880.
  - U. Girish and R. Raz. Eliminating intermediate measurements using pseudorandom generators. In *Proceedings of the 13th Innovations in Theoretical Computer Science Conference, LIPIcs* 215, pages 76:1–76:18, 2022. arXiv:2106.11877.

- A. Ta-Shma. Inverting well conditioned matrices in quantum logspace. In *Proceedings of the 45th Annual ACM Symposium on Theory of Computing*, pages 881–890, 2013.
  - B. Fefferman and C. Y.-Y. Lin. A complete characterization of unitary quantum space. In *Proceedings of the 9th Innovations in Theoretical Computer Science Conference*, LIPIcs 94, pages 4:1–4:21, 2018. arXiv:1604.01384.
10. *Precise quantum computation*. When the gap  $\alpha - \beta$  in  $\text{QMA}(\alpha, \beta)$  is allowed to be exponentially small, rather than inverse-polynomial, the resulting class (known as PreciseQMA) is equal to PSPACE. Correspondingly, a version of the local Hamiltonian problem in which the two energy thresholds are exponentially close is PSPACE-complete. Possible projects include a proof of these facts and a survey of how the complexity of estimating ground-state energies depends on the required precision and on the spectral gap of the Hamiltonian. Background: Chapters 8, 9, and 10.
- B. Fefferman and C. Y.-Y. Lin. Quantum Merlin Arthur with exponentially small gap. arXiv:1601.01975, 2016.
  - Y. Li. A simple proof of PreciseQMA = PSPACE. arXiv:2206.09230, 2022.
  - A. Deshpande, A. V. Gorshkov, and B. Fefferman. The importance of the spectral gap in estimating ground-state energies. *PRX Quantum* 3: 040327, 2022. arXiv:2007.11582.
11. *QMA has perfect completeness*. Whether every problem in QMA has a verification procedure that can be made to accept yes-instances with certainty was a long-standing open question until being recently resolved: it is the case that  $\text{QMA} = \text{QMA}_1 = \text{QMA}(1, 1/2)$ . A possible project is to present a proof of this result. Background: Chapters 8, 9, and 10.
- S. Grewal and D. Rudolph. QMA has perfect completeness. arXiv:2609.13032, 2026.
  - Y. Liu and T. Vidick. Achieving perfect completeness for one- and two-message quantum proof systems. arXiv:2609.15926, 2026.
12. *Quantum 2-SAT and the satisfiability dichotomy*. Classically, 2-SAT is in P while 3-SAT is NP-complete, and an analogous dichotomy is now known for quantum satisfiability (through  $\text{QMA} = \text{QMA}_1$ ). Possible project topics include algorithms for quantum 2-SAT and the hardness of quantum  $k$ -SAT for larger  $k$ . Background: Chapter 10.

- S. Bravyi. Efficient algorithm for a quantum analogue of 2-SAT. *Contemporary Mathematics* 536: 33–48, 2011. arXiv:quant-ph/0602108.
  - I. Arad, M. Santha, A. Sundaram, and S. Zhang. Linear-time algorithm for quantum 2SAT. *Theory of Computing* 14(1): 1–27, 2018. arXiv:1508.06340.
  - N. de Beaudrap and S. Gharibian. A linear time algorithm for quantum 2-SAT. In *Proceedings of the 31st Conference on Computational Complexity, LIPIcs* 50, pages 27:1–27:21, 2016. arXiv:1508.07338.
  - D. Gosset and D. Nagaj. Quantum 3-SAT is QMA<sub>1</sub>-complete. *SIAM Journal on Computing* 45(3): 1080–1128, 2016. arXiv:1302.0290.
13. *Unentangled quantum proofs.* QMA(2) is a variant of QMA in which the verifier receives two quantum witnesses that are promised to be unentangled. While much is unknown about this class, several interesting things are known, and a project could present one or more of these results. Examples include short proofs for NP and the product test and its consequences (including  $\text{QMA}(k) = \text{QMA}(2)$  for  $k \geq 2$  along with error reduction). Background: Chapters 8 and 9.
- H. Kobayashi, K. Matsumoto, and T. Yamakami. Quantum Merlin-Arthur proof systems: Are multiple Merlins more helpful to Arthur? *Chicago Journal of Theoretical Computer Science* 2009: Article 3, 2009. arXiv:quant-ph/0306051.
  - S. Aaronson, S. Beigi, A. Drucker, B. Fefferman, and P. Shor. The power of unentanglement. *Theory of Computing* 5(1): 1–42, 2009. arXiv:0804.0802.
  - H. Blier and A. Tapp. A quantum characterization of NP. *Computational Complexity* 21(3): 499–510, 2012. arXiv:0709.0738.
  - A. W. Harrow and A. Montanaro. Testing product states, quantum Merlin-Arthur games and tensor optimization. *Journal of the ACM* 60(1): 3:1–3:43, 2013. arXiv:1001.0017.
14. *The complexity of detecting entanglement.* Deciding whether a given bipartite density operator is separable or entangled is computationally hard, which connects to the complexity class QMA(2) and its power. Possible project topics include hardness results for this problem. Background: Chapter 8.
- L. Gurvits. Classical complexity and quantum entanglement. *Journal of Computer and System Sciences* 69(3): 448–484, 2004. arXiv:quant-ph/0303055.
  - L. M. Ioannou. Computational complexity of the quantum separability problem. *Quantum Information and Computation* 7(4): 335–370, 2007. arXiv:quant-ph/0603199.

- S. Gharibian. Strong NP-hardness of the quantum separability problem. *Quantum Information and Computation* 10(3&4): 343–360, 2010. arXiv:0810.4507.
  - A. W. Harrow and A. Montanaro. Testing product states, quantum Merlin-Arthur games and tensor optimization. *Journal of the ACM* 60(1): 3:1–3:43, 2013. arXiv:1001.0017.
15. *Quantum advice*. Chapter 11 proves that  $\text{BQP}/\text{qpoly} \subseteq \text{PP}/\text{poly}$ . A project could go further and cover other known results connected with quantum advice. Background: Chapter 11.
- S. Aaronson. Limitations of quantum advice and one-way communication. *Theory of Computing* 1(1): 1–28, 2005. arXiv:quant-ph/0402095.
  - S. Aaronson and G. Kuperberg. Quantum versus classical proofs and advice. *Theory of Computing* 3(1): 129–157, 2007. arXiv:quant-ph/0604056.
  - R. Raz. Quantum information and the PCP theorem. *Algorithmica* 55(3): 462–489, 2009. arXiv:quant-ph/0504075.
  - S. Aaronson and A. Drucker. A full characterization of quantum advice. *SIAM Journal on Computing* 43(3): 1131–1183, 2014. arXiv:1004.0377.
  - J. Bostanci, A. Huang, and V. Vaikuntanathan. Separating quantum and classical advice with good codes. arXiv:2602.09385, 2026.
16. *Stoquastic Hamiltonians*. A stoquastic Hamiltonian is one whose off-diagonal entries are all non-positive. Hamiltonians like this arise in physics, and are special because they do not incur the so-called sign problem in quantum Monte Carlo methods. The local Hamiltonian problem for stoquastic Hamiltonians is complete for a class  $\text{StoqMA}$  satisfying  $\text{MA} \subseteq \text{StoqMA} \subseteq \text{AM} \cap \text{QMA}$ , while a frustration-free version of the problem is  $\text{MA}$ -complete. A project could present these results and survey what is known about  $\text{StoqMA}$ . Background: Chapters 8 and 10.
- S. Bravyi, A. J. Bessen, and B. M. Terhal. Merlin-Arthur games and stoquastic complexity. arXiv:quant-ph/0611021, 2006.
  - S. Bravyi, D. P. DiVincenzo, R. I. Oliveira, and B. M. Terhal. The complexity of stoquastic local Hamiltonian problems. *Quantum Information and Computation* 8(5): 361–385, 2008. arXiv:quant-ph/0606140.
  - S. Bravyi and B. M. Terhal. Complexity of stoquastic frustration-free Hamiltonians. *SIAM Journal on Computing* 39(4): 1462–1485, 2009. arXiv:0806.1746.

17. *Commuting local Hamiltonians.* Much remains unknown about the complexity of the local Hamiltonian problem when the terms of the Hamiltonian are required to commute with one another. For 2-local commuting Hamiltonians the problem is in NP, by an argument based on the structure of finite-dimensional  $C^*$ -algebras, and some 3-local cases and certain lattice models are also known to be in NP. A project could present one or more of these results and survey what remains unanswered. Background: Chapter 10.
  - S. Bravyi and M. Vyalyi. Commutative version of the local Hamiltonian problem and common eigenspace problem. *Quantum Information and Computation* 5(3): 187–215, 2005. arXiv:quant-ph/0308021.
  - D. Aharonov and L. Eldar. On the complexity of commuting local Hamiltonians, and tight conditions for topological order in such systems. In *Proceedings of the 52nd Annual IEEE Symposium on Foundations of Computer Science*, pages 334–343, 2011. arXiv:1102.0770.
  - N. Schuch. Complexity of commuting Hamiltonians on a square lattice of qubits. *Quantum Information and Computation* 11(11–12): 901–912, 2011. arXiv:1105.2843.
18. *The guided local Hamiltonian problem.* Some proposals for quantum advantage in quantum chemistry involve preparing an approximation to a ground state and then running phase estimation. The guided local Hamiltonian problem formalizes this idea: a local Hamiltonian is given together with a classical description of a state having inverse-polynomial overlap with its ground space. This problem is BQP-complete when the ground energy must be estimated to inverse-polynomial precision, but it admits an efficient classical algorithm at constant precision. A project could present these results and discuss what they suggest about prospects for quantum advantage. Background: Chapter 10.
  - S. Gharibian and F. Le Gall. Dequantizing the quantum singular value transformation: hardness and applications to quantum chemistry and the quantum PCP conjecture. *SIAM Journal on Computing* 52(4): 1009–1038, 2023. arXiv:2111.09079.
  - C. Cade, M. Folkertsma, S. Gharibian, R. Hayakawa, F. Le Gall, T. Morimae, and J. Weggemans. Improved hardness results for the guided local Hamiltonian problem. In *Proceedings of the 50th International Colloquium on Automata, Languages, and Programming*, LIPIcs 261, pages 32:1–32:19, 2023. arXiv:2207.10250.
19. *Adiabatic quantum computation.* Adiabatic quantum computation is a model in which a computation is performed by slowly varying a Hamiltonian, start-

ing from one whose ground state can be easily prepared and ending with one whose ground state encodes the answer to the problem. This model is polynomially equivalent to the quantum circuit model. Possible projects include a proof of this equivalence, which makes use of the clock construction of Chapter 10. Background: Chapter 10.

- D. Aharonov, W. van Dam, J. Kempe, Z. Landau, S. Lloyd, and O. Regev. Adiabatic quantum computation is equivalent to standard quantum computation. *SIAM Journal on Computing* 37(1): 166–194, 2007. arXiv:quant-ph/0405098.
20. *Quantum zero-knowledge*. Zero-knowledge must be defined with some care when the verifier is quantum, as standard classical techniques (e.g., rewinding the verifier) do not work directly against quantum verifiers. Possible projects include the quantum rewinding lemma and its applications, zero-knowledge proofs for NP that are secure against quantum verifiers, zero-knowledge proofs for QMA, and general properties of quantum zero-knowledge proof systems. Background: Chapters 8 and 10, and Part IV.
- T. Vidick and J. Watrous. Quantum proofs. *Foundations and Trends in Theoretical Computer Science* 11(1–2): 1–215, 2016. arXiv:1610.01664.
  - J. Watrous. Zero-knowledge against quantum attacks. *SIAM Journal on Computing* 39(1): 25–58, 2009. arXiv:quant-ph/0511020.
  - H. Kobayashi. General properties of quantum zero-knowledge proofs. In *Proceedings of the Fifth Theory of Cryptography Conference, Lecture Notes in Computer Science* 4948, pages 107–124, 2008. arXiv:0705.1129.
  - S. Hallgren, A. Kolla, P. Sen, and S. Zhang. Making classical honest verifier zero knowledge protocols secure against quantum attacks. In *Proceedings of the 35th International Colloquium on Automata, Languages and Programming, Lecture Notes in Computer Science* 5126, pages 592–603, 2008.
  - A. Broadbent, Z. Ji, F. Song, and J. Watrous. Zero-knowledge proof systems for QMA. *SIAM Journal on Computing* 49(2): 245–283, 2020. arXiv:1604.02804.
21. *Quantum statistical zero-knowledge*. The class QSZK consists of promise problems having quantum interactive proof systems that are zero-knowledge in a statistical sense, without cryptographic assumptions. A natural problem concerning the distinguishability of quantum states is complete for this class, and several properties of QSZK follow — including closure under complementation and containment in PSPACE. A project could present these results. Background: Chapters 5 and 8.

- T. Vidick and J. Watrous. Quantum proofs. *Foundations and Trends in Theoretical Computer Science* 11(1–2): 1–215, 2016. arXiv:1610.01664.
  - A. Sahai and S. Vadhan. A complete problem for statistical zero knowledge. *Journal of the ACM* 50(2): 196–249, 2003.
  - J. Watrous. Limits on the power of quantum statistical zero-knowledge. In *Proceedings of the 43rd Annual IEEE Symposium on Foundations of Computer Science*, pages 459–468, 2002. arXiv:quant-ph/0202111.
  - J. Watrous. Zero-knowledge against quantum attacks. *SIAM Journal on Computing* 39(1): 25–58, 2009. arXiv:quant-ph/0511020.
22. *Density matrix consistency and zero-knowledge proofs for QMA.* The consistency of local density matrices problem asks whether there exists a global state consistent with a given collection of reduced states on small subsets of qubits. This problem is QMA-complete, and the proof of its QMA-hardness under Karp reductions rests on history states whose reduced states can be computed without knowledge of the witness. The same idea leads to zero-knowledge proof systems for QMA. A project could present one or both of these results. Background: Chapter 10.
- A. Broadbent and A. B. Grilo. QMA-hardness of consistency of local density matrices with applications to quantum zero-knowledge. *SIAM Journal on Computing* 51(4): 1400–1450, 2022. arXiv:1911.07782.
  - Y.-K. Liu. Consistency of local density matrices is QMA-complete. In *Approximation, Randomization, and Combinatorial Optimization, Lecture Notes in Computer Science* 4110, pages 438–449, 2006. arXiv:quant-ph/0604166.
  - A. B. Grilo, W. Slofstra, and H. Yuen. Perfect zero knowledge for quantum multiprover interactive proofs. In *Proceedings of the 60th Annual IEEE Symposium on Foundations of Computer Science*, pages 611–635, 2019. arXiv:1905.11280.
23. *Quantum Arthur–Merlin games.* In a quantum Arthur–Merlin game, the verifier’s messages to the prover are uniformly random bits. Such games with three messages are as powerful as general quantum interactive proof systems, while those with two messages are contained in  $\text{BP} \cdot \text{PP}$  — in contrast with the classical case, where any constant number of messages can be reduced to two. A project could present these results and related work. Background: Chapters 8 and 9, and Part IV.
- C. Marriott and J. Watrous. Quantum Arthur–Merlin games. *Computational Complexity* 14(2): 122–152, 2005. arXiv:cs/0506068.

- H. Kobayashi, F. Le Gall, and H. Nishimura. Generalized quantum Arthur–Merlin games. *SIAM Journal on Computing* 48(3): 865–902, 2019. arXiv:1312.4673.
  - Y. Liu and T. Vidick. Achieving perfect completeness for one- and two-message quantum proof systems. arXiv:2609.15926, 2026.
24. *Distinguishing quantum circuits.* A natural computational problem concerning quantum circuits is to decide whether the channels implemented by two given circuits are close together or far apart. This problem is QIP-complete, and restricted variants of it are complete for other classes, including QMA and QSZK. A project could present these results. Background: Chapter 8 and Part IV.
- B. Rosgen and J. Watrous. On the hardness of distinguishing mixed-state quantum computations. In *Proceedings of the 20th Annual IEEE Conference on Computational Complexity*, pages 344–354, 2005. arXiv:cs/0407056.
  - T. Vidick and J. Watrous. Quantum proofs. *Foundations and Trends in Theoretical Computer Science* 11(1–2): 1–215, 2016. arXiv:1610.01664.
  - D. Janzing, P. Wocjan, and T. Beth. “Non-identity-check” is QMA-complete. *International Journal of Quantum Information* 3(3): 463–473, 2005. arXiv:quant-ph/0305050.
  - B. Rosgen. Distinguishing short quantum computations. In *Proceedings of the 25th International Symposium on Theoretical Aspects of Computer Science*, LIPIcs 1, pages 597–608, 2008. arXiv:0712.2595.
25. *Quantum refereed games.* In a refereed game, a verifier interacts with two competing provers, one trying to convince it to accept and the other to reject. Quantum refereed games characterize EXP, just as classical refereed games do, while those with a single round of messages to each prover characterize PSPACE. A project could present these results, which rest on semidefinite programming and parallel algorithms for min-max problems. Background: Part IV.
- G. Gutoski and J. Watrous. Toward a general theory of quantum games. In *Proceedings of the 39th Annual ACM Symposium on Theory of Computing*, pages 565–574, 2007. arXiv:quant-ph/0611234.
  - R. Jain and J. Watrous. Parallel approximation of non-interactive zero-sum quantum games. In *Proceedings of the 24th Annual IEEE Conference on Computational Complexity*, pages 243–253, 2009. arXiv:0808.2775.

- G. Gutoski and X. Wu. Parallel approximation of min-max problems. *Computational Complexity* 22(2): 385–428, 2013. arXiv:1011.2787.
26. *Classical verification of quantum computation.* A natural question is whether a classical verifier can check the result of a quantum computation carried out by an untrusted quantum prover. Mahadev’s protocol shows that this is possible, under the assumption that the learning with errors problem is hard for quantum computers. A project could explain how the protocol works, treating the cryptographic components as black boxes if desired. Background: Chapter 10.
- U. Mahadev. Classical verification of quantum computations. *SIAM Journal on Computing* 51(4): 1172–1229, 2022. arXiv:1804.01082.
  - T. Vidick. Verifying quantum computations at scale: a cryptographic leash on quantum devices. *Bulletin of the American Mathematical Society* 57(1): 39–76, 2020.
27. *Something else.* If you have an idea of your own for a topic connected with quantum complexity theory, you are welcome to pursue it — but please confirm it with me first!